MEETHILA GADE

Computational Biologist | Single-Cell Multi-omics | Precision Medicine & Target Discovery
Email: meethilagade@gmail.com | Phone: 917-673-9459 | Website: brain-discourse.github.io
LinkedIn: linkedin.com/in/meethilagade | Github: github.com/brain-discourse

Computational biologist with integrated wet-lab and dry-lab training, specializing in single-nucleus multi-omics, clinical genomics, and translational biomarker discovery. My work integrates variant discovery, genotype-resolved transcriptomics, and scalable NGS pipeline development to characterize disease-associated cell states and identify clinically actionable genetic findings across large patient cohorts.

TECHNICAL SKILLS

- **Programming:** R, Python, Bash, SQL
- **Statistical modeling:** Mixed-effects and hierarchical models; covariate-adjusted association testing; classification and regression modeling (lme4, scikit-learn); variant burden testing
- **Transcriptomics:** Bulk and scRNA-seq/ATAC-seq; alignment (STAR, kallisto, WASP, CellRanger); QC; data integration/batch correction; clustering and cell-type annotation (Seurat, Scanpy, popV); RNA velocity; differential expression and abundance analysis (limma, edgeR, DESeq2, MAST, dreamlet); pathway and gene set enrichment
- **Multi-Omics:** genotype-expression linkage; donor/batch-aware pseudobulk analysis; cell-state and gene program discovery using variational inference (scVI-tools, scANVI, WGCNA); gene regulatory network inference (SCENIC)
- **Genomic analysis:** WES/WGS/Duplex sequencing processing; somatic/germline variant calling; CNV/LOH/SV identification; annotation and variant prioritization (GATK, Mutect2, BWA, MoChA, ANNOVAR)
- **Compute & reproducibility:** HPC/Slurm, containerized environments, workflow orchestration, reproducible reporting, and cloud fundamentals (AWS S3, Batch; Git, Conda, Singularity, Nextflow, R Shiny)
- **Experimental:** Nuclei isolation, Smart-seq2, NGS library preparation, WGA/PTA/MDA/PicoPLEX, PCR/TaqMan, digital PCR, flow cytometry, tissue dissociation, mammalian cell culture, immunofluorescence/IHC, confocal microscopy, experimental design, ADME/PK assay optimization and modeling (NLME, Phoenix WinNonlin, NONMEM PBPK)

CORE RESEARCH CONTRIBUTIONS

SoMoSeq: Resolving somatic variant-to-expression linkage at single-cell resolution in mosaic human tissue

- Developed SoMoSeq, a multi-omic single nucleus DNA + RNA approach to link variant status with transcriptional profiles from the same nucleus, enabling genotype-to-expression analysis in mosaic human tissue.
- Built snRNA-seq workflows for genotype-resolved analysis across 4,800 processed nuclei (1,800 sequenced), including QC, alignment, normalization, integration, clustering, differential expression, and pathway analysis.
- Developed reproducible HPC-based analysis infrastructure for scalable evaluation of clustering, normalization, and integration strategies across tools and datasets.

Cell-type-specific transcriptional consequences of somatic variants: autonomous vs. non-cell-autonomous effects

- Identified variant-associated transcriptional changes using donor-aware mixed-effects models and single-cell differential expression frameworks with covariate adjustment, distinguishing responses in variant-bearing cells from tissue microenvironment-associated changes in neighboring variant-negative cells across cell types.

Variant prioritization and diagnostic biomarker discovery in clinical disease cohorts

- Drove somatic/germline variant and CNV discovery across 300+ clinical cases through scalable NGS pipelines and pathogenic variant prioritization.
- Integrated functional, clinical, and population-level datasets (gnomAD, OMIM, ClinVar) with multi-caller concordance analysis to support variant interpretation.
- Evaluated sensitivity and specificity tradeoffs of single-nucleus WGA modalities (PTA/MDA) for variant detection using a unified Nextflow-based workflow (PTATO) to quantify genome-wide allelic dropout and amplification bias, informing assay selection for low-frequency variant analysis.

RESEARCH EXPERIENCE

University of North Carolina at Chapel Hill, NC | Graduate Research Assistant 2020-2026

- Integrated experimental and computational omics approaches to study how mosaic genotype influences cell state and disease phenotypes in human brain tissue.
- Designed and executed bulk and single-cell sequencing assays, including WGS/WES, duplex sequencing, Smart-seq2, WGA/PTA/MDA, and SoMoSeq; led downstream CNV/LOH, single-nucleus WGS, and snRNA-seq analyses from study design through biological interpretation.

- Developed reproducible HPC-based bioinformatics workflows for large-scale RNA-seq and NGS analysis, incorporating batch correction, technical artifact assessment, and model diagnostics to preserve biology across heterogeneous samples.
- Partnered with clinical and experimental teams across a 300+ case cohort to refine study design and integrate genomic findings with clinical phenotypes for variant prioritization, interpretation and validation, improving diagnostic yield in previously unresolved cases.

Biogen, Boston, MA | Computational Biology Intern — Summer 2023

- Established standardized cell-type annotation nomenclature and QC thresholds across 4 neurodegenerative disease programs (ALS, AD, PD, MS; over 3.5 million cells), improving cross-disease annotation consistency, interpretability, and enabling consistent reuse of 10x Genomics scRNA-seq outputs by 3+ therapeutic-area teams.
- Integrated large-scale public references (Allen Brain Atlas, CELLxGENE, GTEx) with internal scRNA-seq datasets to support a comprehensive CNS reference atlas, enabling cross-disease cell-state and pathway analyses for translational biomarker discovery.

Columbia University Medical Center, NYC | Graduate Research Assistant — 2018- 2020

- Analyzed longitudinal transcriptomic time-series data (n=25 mice), to identify early molecular programs associated with neuromuscular degeneration and aging-related functional decline.
- Evaluated neurotoxic effects of heavy metals and organophosphates in iPSC-derived motor neurons and ALS mouse models using dose-response, immunostaining/immunofluorescence, and behavioral phenotyping assays.

Regeneron, Tarrytown, NY | Graduate Intern, VelociGene — Summer 2019

- Integrated transcriptomic and imaging-derived phenotypic data from CRISPR-edited cellular models in a cisplatin-induced ototoxicity screen, prioritizing candidate genes associated with toxicity response and supporting target evaluation for translational follow-up studies.

Additional Research Experience

- CDRI/DKFZ: comparative genomics (2017-2018); JRF Global: ADME/PK assay development (Summer 2018).

SELECTED PUBLICATIONS (*Google Scholar*)

- **Brain (2022)** Lai, D., **Gade, M. (co-first)**, et al. Somatic variants in diverse genes lead to a spectrum of focal cortical malformations.
- **Nature Genetics (2023)** Miller, K.E., Rivaldi, A.C., Shinagawa, N., **Gade, M.**, et al. Post-zygotic rescue of meiotic errors causes brain mosaicism and focal epilepsy.
- **J Cachexia Sarcopenia Muscle (2023)** Comfort, N., **Gade, M.**, et al. Transcriptomic analysis of aging mouse sciatic nerve reveals early pathways leading to sarcopenia.
- **Gade, M.**, et al. SoMoSeq: A method for genotype-informed single-nucleus RNA-seq of mosaic brain tissue. **(Manuscript in preparation)**

SELECTED PLATFORM PRESENTATIONS

- **ASHG 2024 (Denver, CO):** Genotype-Informed scRNA-Seq Reveals Somatic Loss of Heterozygosity in Hemimegalencephaly with *PIK3CA* Mutations.
- **ASHG 2023 (Washington, DC):** SoMoSeq: A method for genotype informed scRNA-seq of mosaic brain tissue.

EDUCATION

- **University of North Carolina at Chapel Hill - Ph.D., Pharmaceutical Sciences** — May 2026
  Thesis: Somatic Mosaicism in Human Brain - Variant Discovery, Genotype-Resolved Single-Nucleus Multi-omics, and Cell-Type-Specific Transcriptional Consequences.
- **Columbia University - M.P.H., Environmental Health Sciences; Advanced Certificate, Toxicology** — 2020
- **St. Xavier's College - Post Graduate Diploma, Clinical Research** — 2018
- **University of Delhi - B.Sc., Biology & Chemistry** — 2017

AWARDS AND TRAINING

- American Epilepsy Society - Predoctoral Fellowship — *2024*
- Cold Spring Harbor Laboratory - Single Cell Analysis (Regeneron Scholars Program) — *2022*